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UNIVERSITI TEKNOLOGI MALAYSIA
Faculty of Artificial Intelligence

TOPIC 3:
DEALING WITH DUPLICATES AND OUTLIERS

DATA WRANGLING AND VISUALIZATION (SAIA2143)

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COURSE LEARNING OUTCOMES

At the end of this topic, students should be able to:

i. **Define and explain** the concept of duplicates and outliers in a dataset.

ii. **Identify common causes** of duplicate data entries and outliers during data collection and processing.

iii. **Apply Excel tools and functions** (e.g., Remove Duplicates, COUNTIF, Conditional Formatting) to detect and manage duplicates.

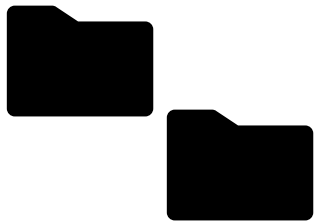
iv. **Use statistical techniques** (e.g., Z-Score and Interquartile Range [IQR]) to detect outliers in a dataset.

v. **Interpret the impact** of duplicates and outliers on data analysis and decision-making.

What are Duplicates?

- Definition – **Duplicates** refer to rows or records in a dataset that appear more than once with exactly the same values in one or more columns.

- These redundant entries can skew data analysis, lead to inaccurate conclusions, and increase the size of the dataset unnecessarily.



Example of Duplicate Scenario



We must ensure that we have a single source of truth for each observation. There are scenarios where we have multiple observations about a single customer, either because there is a double form submission by the user, or errors were made during data entry. Regardless of how this comes about, we need to develop a single, complete version of the truth.

This will help businesses ensure that:

1. Time and money is not wasted targeting the same customer twice.
2. A trusted and complete version of the truth is developed, and reliable business decisions can be made based on the analysis results.

Example of Duplicate in Excel

ID	Name	Email
001	Alice Tan	alice@email.com
002	John Lim	john@email.com
003	Alice Tan	alice@email.com
004	Sarah Wong	sarah@email.com

Row 1 and Row 3 are **duplicates** because **Name** and **Email** are exactly the same.

Causes of Duplicate Data

Num	Cause	Explanation	Example
1	**Human Error**	Manual entry may result in typing the same record more than once.	“Amirul Azmi” typed twice by mistake.
2	**Copy-Paste Errors**	Data copied from other sheets may include repeated entries.	Pasting order records without checking for existing data.
3	**Multiple Data Sources**	Same data pulled from different systems can overlap.	Customer info from CRM and eCommerce tools may both list “John Lee.”
4	**Lack of Unique Identifier**	No unique ID makes it hard to distinguish real duplicates.	Two entries for “Ahmad Ali” might not be the same person.
5	**Incomplete Data Cleaning**	Extra spaces or inconsistent formats can hide duplicates.	“alia@gmail.com” ≠ “alia@gmail.com ” (trailing space).
6	**System Sync Issues**	Delayed or faulty data syncs between platforms may double the data.	A form submission duplicated due to a page reload.

Problems Caused by Duplicates

Problem	Explanation	Example
Inflated counts	Duplicate entries artificially increase the number of records, leading to incorrect totals.	If a customer appears twice, the customer count is overstated.
Skewed analysis	Analysis using duplicated data can produce misleading insights or trends.	A duplicated sale may make average sales figures appear higher than they truly are.
Poor decision-making	Decisions based on incorrect data may result in wasted resources or flawed strategies.	Marketing may target the same person multiple times, wasting time and budget.

Detecting Duplicates in Excel

1	Using **Remove Duplicates**
2	Using **COUNTIF()**
3	Using **Conditional Formatting** to highlight duplicates – Refer to Chapter 2

1. Remove Duplicates

- It checks for rows (or selected columns) with the same values and deletes the **extra copies**, leaving only the first instance.

- **When would you use it?**
  a. Cleaning customer or product lists.
  b. Removing repeated entries in a survey.
  c. Filtering unique values for analysis or reporting.

Let's Try Together

1. Removing Duplicates in Excel

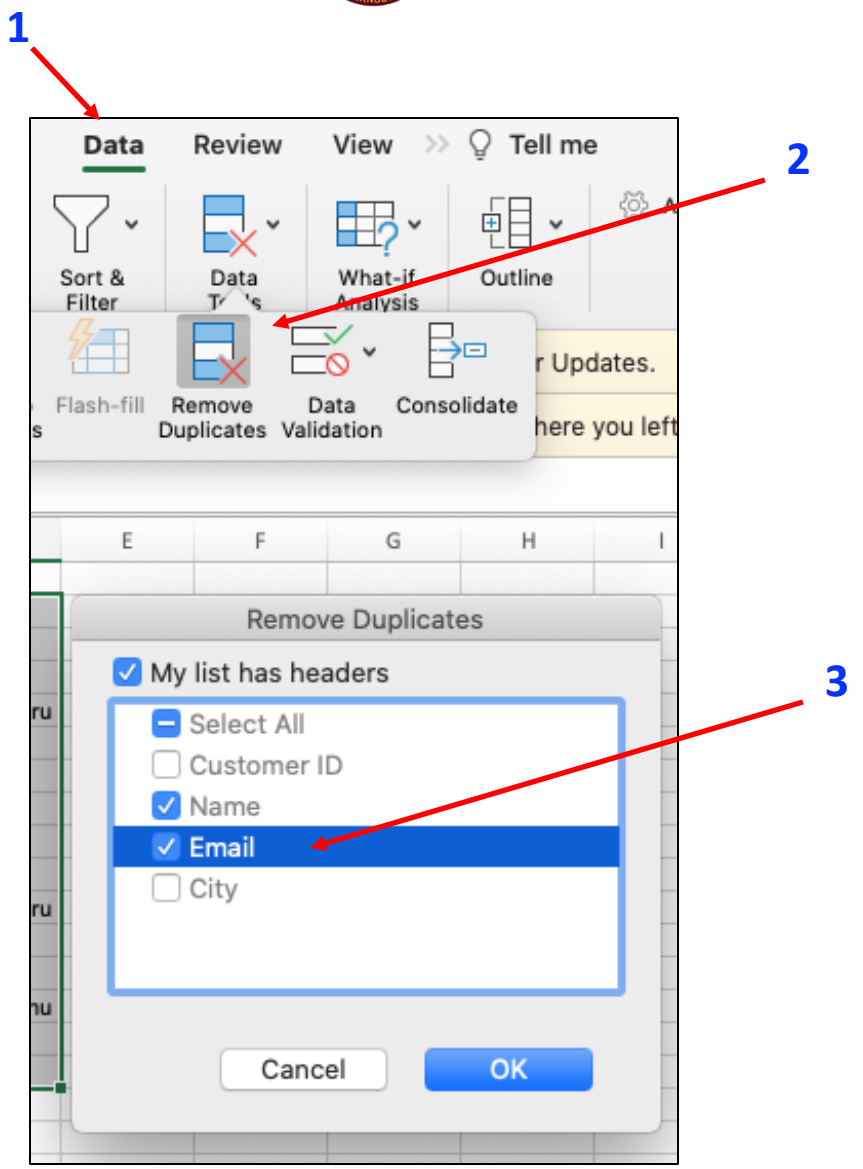
Scenario

You work for a marketing team managing a customer email list. You've noticed that some customers are listed more than once, and this might lead to multiple emails being sent to the same person. Your task is to clean the list by removing duplicate entries.

Data

	A	B	C	D
1	**Customer ID**	**Name**	**Email**	**City**
2	C001	Alia Omar	alia@gmail.com	Penang
3	C002	Jason Low	jason@hotmail.com	KL
4	C003	Alia Omar	alia@gmail.com	Penang
5	C004	Nadira Yusuf	nadira@yahoo.com	Johor Bahru
6	C005	Jason Low	jason@hotmail.com	KL
7	C006	Farid Rahman	farid@gmail.com	Melaka
8	C007	Aina Ismail	aina@yahoo.com	Penang
9	C008	Farid Rahman	farid@gmail.com	Melaka
10	C009	John Lee	john.lee@gmail.com	KL
11	C010	Nadira Yusuf	nadira@yahoo.com	Johor Bahru
12	C011	Zara Ibrahim	zara.ibrahim@gmail.com	Perak
13	C012	Aina Ismail	aina@yahoo.com	Penang
14	C013	Amirul Azmi	amirul@gmail.com	Terengganu
15	C014	John Lee	john.lee@gmail.com	KL
16	C015	Zara Ibrahim	zara.ibrahim@gmail.com	Perak

1. **Open the data in Excel**.
2. Select the range (including headers).
3. Go to the **Data** tab → click on **Remove Duplicates**.
4. In the dialog box, tick the columns you want to check for duplicates:
    - For this activity, choose: **Name** and **Email**.
5. Click **OK** and Excel will remove the duplicate rows.



Output

	A	B	C	D
1	**Customer ID**	**Name**	**Email**	**City**
2	C001	Alia Omar	alia@gmail.com	Penang
3	C002	Jason Low	jason@hotmail.com	KL
4	C004	Nadira Yusuf	nadira@yahoo.com	Johor Bahru
5	C006	Farid Rahman	farid@gmail.com	Melaka
6	C007	Aina Ismail	aina@yahoo.com	Penang
7	C009	John Lee	john.lee@gmail.com	KL
8	C011	Zara Ibrahim	zara.ibrahim@gmail.com	Perak
9	C013	Amirul Azmi	amirul@gmail.com	Terengganu



2. COUNTIF()

- The COUNTIF function in Excel is used to **count the number of cells that meet a specific condition (or criteria)**.

- Syntax: COUNTIF(range, criteria)

The range of cells you want to check. The condition you want to apply.

- **Common Use Cases:**
  a. Identifying duplicates.
  b. Counting how many times a name or item appears.
  c. Filtering data by condition (e.g., "greater than", "less than", "equal to").
  d. Basic data analysis or summaries in dashboards.

Examples of COUNTIF()

Scenario	Excel formula
Count how many times "Alice" appears in a list	=COUNTIF(A2:A10, "Alice")
Counts all numbers in B2:B20 that are greater than 100	=COUNTIF(B2:B20, ">100")
Count blank cells	=COUNTIF(A2:A10, "")

CLASS ACTIVITY
2. COUNTIF()

Data

Scenario: Employee Attendance Tracker

You are managing attendance records for a company. Employees check in every day, and their names are recorded. Sometimes, due to a system glitch, the same employee might get recorded **more than once a day**. You want to identify **duplicate entries**.

	A	B
1	**Employee Name**	**Output**
2	Alice	
3	Bob	
4	Charlie	
5	Alice	
6	Diana	
7	Bob	
8	Edward	
9	Alice	

Output

Formula:

=IF(COUNTIF(A2:A9) – Press **F4** to lock the cell references with dollar sign

=IF(COUNTIF($A$2:$A$9, A2)>1, "Duplicate", "Unique")

1	Employee Name	Output
2	Alice	Duplicate
3	Bob	Duplicate
4	Charlie	Unique
5	Alice	Duplicate
6	Diana	Unique
7	Bob	Duplicate
8	Edward	Unique

CLASS EXERCISE

Scenario: Customer Orders Analysis

You're analyzing customer order data for an online store.

1. See how many times each customer placed an order.
2. Identify frequent customers (those who ordered more than once).

Data

	A	B	C
1	Customer Name	Number of order per customer	Frequent/One-time
2	Sarah		
3	James		
4	Michael		
5	Sarah		
6	Emily		
7	David		
8	Michael		
9	Sarah		
10	James		
11	Olivia		
12	Emily		
13	Michael		
14	Noah		
15	Emily		
16	Sarah		

RECALL: Conditional Formatting

- Allows users to automatically apply formatting (e.g., colors, bold text, icons) to cells based on specific conditions or criteria.
- It helps visually highlight key data trends, anomalies, or outliers.
- Types of conditional formatting:
  1. Highlight cells rules
  2. Top/bottom rules
  3. Color scales
  4. Data bars
  5. Icon sets

	A	B	C	D	E	F
1						
2		**Month**	**Sales**			**Sales Target**
3		January		$ 35 600	✓	$ 30 000
4		February		$ 52 161	✓	$ 36 000
5		March		$ 40 164	✓	$ 36 000
6		April		$ 41 077	✓	$ 38 000
7		May		$ 40 425	✓	$ 40 000
8		June		$ 36 513	✗	$ 40 000
9		July		$ 38 730	✗	$ 43 000
10		August		$ 47 076	✓	$ 45 000
11		September		$ 43 946	✗	$ 45 000
12		October		$ 50 596	✓	$ 47 000
13		November		$ 51 509	✓	$ 50 000
14		December		$ 35 992	✗	$ 50 000
16		**Total**		**$ 513 790**	✓	**$ 500 000**



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Scenario: Duplicate Entries in Customer Records

You are working in a customer service department. After importing contact data from two different sources, you suspect that **some customers appear more than once**. Your goal is to identify these **duplicate names** to clean up the list using **Conditional Formatting**.

Data

	A	B
1	Customer Name	Email
2	Alice Tan	alice@example.com
3	Brian Lee	brian@example.com
4	Catherine Wong	cathy@example.com
5	Alice Tan	alice.tan@sample.com
6	Daniel Ng	daniel@example.com
7	Brian Lee	brianlee@sample.com
8	Ethan Ho	ethan@example.com
9	Catherine Wong	wong.c@sample.com
10	Alice Tan	at_customer@demo.com
11	Brian Lee	brian.sales@company.com
12	Catherine Wong	cwong@ecom.com
13	Fiona Goh	fiona@example.com
14	Ethan Ho	eho@customer.org

What are Outliers?

- An **outlier** is a data point that significantly differs from other values in a dataset.
- It may be much higher or lower than the majority of the data.
- Outliers can occur due to errors, variability, or unusual conditions — and they can **skew analysis** if not handled properly.

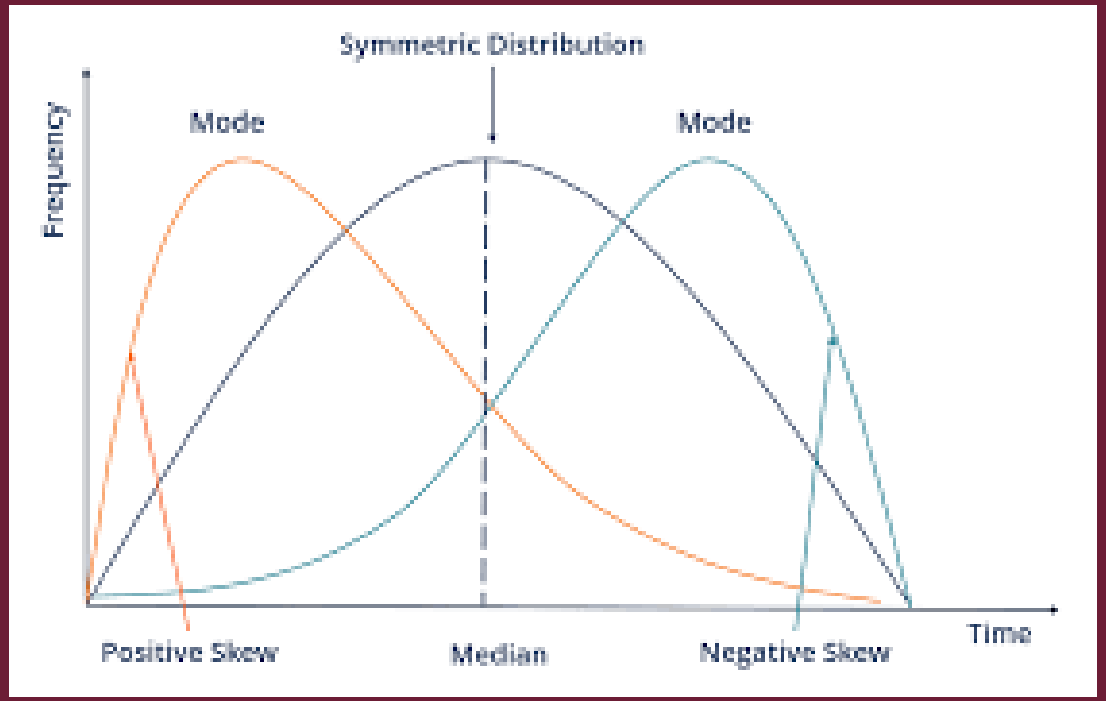
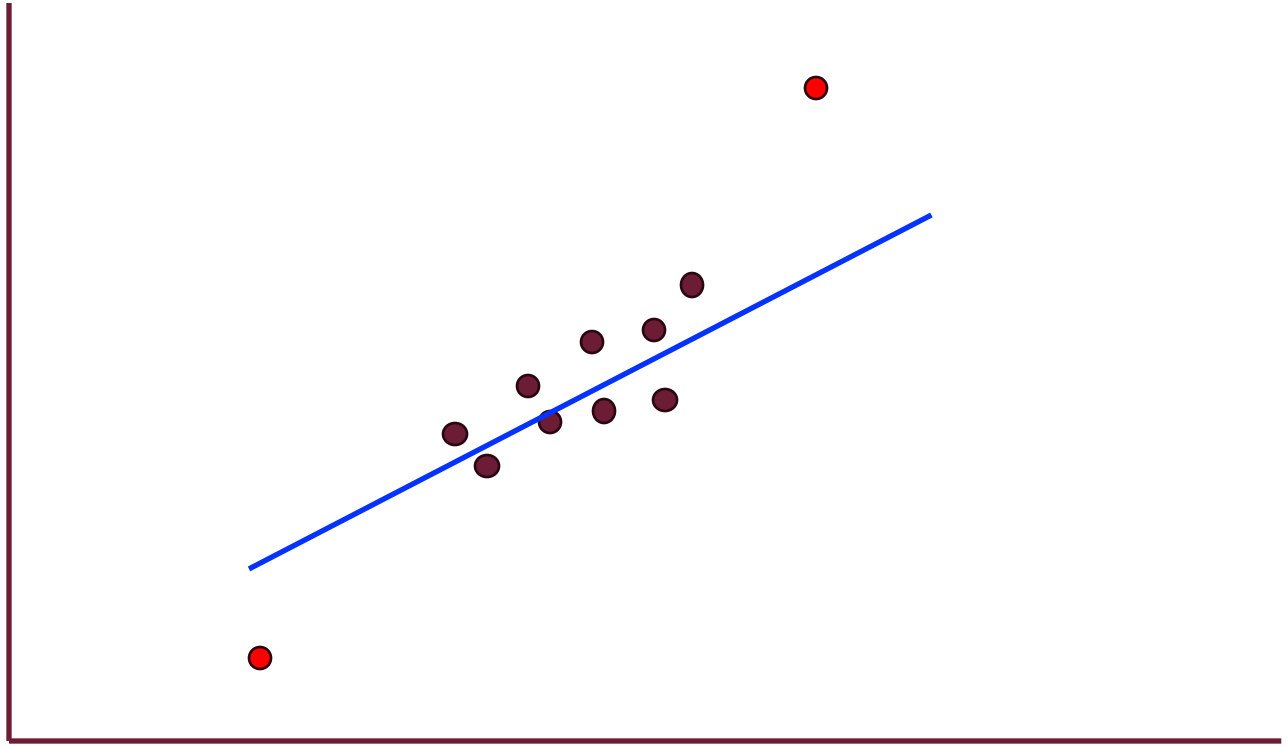


Illustration of Outliers



Numbers that are far greater or far less than the data available, and these numbers can affect statistical outcomes, even skewing them at times.

Common Causes of Outliers

Cause	Description	Example
1. Data Entry Errors	Mistakes during manual input or copying data	Entering 5000 instead of 500 in sales
2. Measurement Errors	Errors from faulty equipment or systems	Sensor records 120°C when it should be below 50°C
3. Data Processing Errors	Issues during import/export, formatting, or merging datasets	A text field interpreted as a number, or vice versa
4. Natural Variation	Valid, rare events in real-world behavior	One-off viral sales event increases figures significantly
5. Sampling Errors	Data from a biased or unrepresentative sample	Only surveying top customers in a market study
6. Fraud or Manipulation	Intentionally manipulated or falsified data	Faked numbers in financial or performance reports
7. System or Policy Changes	Changes in business rules or system behavior that shift data patterns	A promo launch causing a temporary spike in website traffic or sales

Detecting Outliers in Excel

Technique	Description	Example Formula/Action
1. Sorting	Sort values in ascending/descending order to visually spot extreme values.	Use **Sort** in the **Data tab**
2. Conditional Formatting	Highlight values above/below a threshold.	Use **Highlight Cell Rules** → **Greater Than / Less Than**
3. Z-Score (Standard Deviation)	Values more than 2 or 3 standard deviations from the mean are potential outliers.	=ABS((A2-AVERAGE(A:A))/STDEV.P(A:A)) > 2
4. IQR Method (Interquartile Range)	Identify values outside Q1 - 1.5×IQR and Q3 + 1.5×IQR.	Use QUARTILE() and IQR = Q3 - Q1
5. Box and Whisker Chart	Visual method to spot outliers using Excel's chart tools.	Insert → **Box & Whisker Chart** (Excel 2016+)
6. FILTER or IF Logic	Create a custom rule to isolate values outside expected range.	=IF(A2>UpperLimit,"Outlier","")
7. Power Query	Use Power Query to group, filter, and flag unusual values in large datasets.	Data → **Get & Transform**

Z-Score (Standard Deviation)

- A **Z-Score** (also known as a **standard score**) tells you **how far a data point is from the mean** (average) of a dataset, **in terms of standard deviations**.

- Z-scores help:

a. Understand how unusual a value is

b. Detect outliers

c. Standardize values for comparison

Formula:

$$\text{Z-score} = \frac{(X - \mu)}{\sigma}$$

Where:

- X = your data point
- μ = the **mean** (average)
- σ = the **standard deviation**

Example

Let's say we have test scores: **[70, 75, 80, 85, 90]**

- Mean (μ) = 80
- Standard Deviation (σ) ≈ 7.91

Now, we calculate the Z-score for **90**:

$$Z=\frac{(90-80)}{7.91} \approx 1.26$$

➡ This means 90 is **1.26 standard deviations above** the average.

Interpreting Z-scores

Z-Score	Interpretation
0	Exactly average
1	1 SD above the mean
-1	1 SD below the mean
> +2 or < -2	Possibly an outlier

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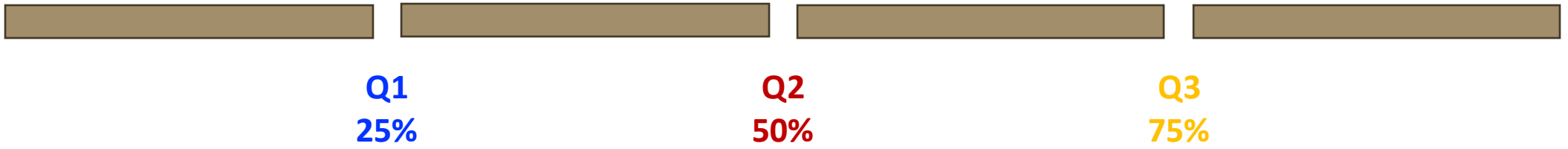
Scenario:

- You are analyzing the monthly sales figures (in RM) for 10 branches of a retail store to detect any unusual sales patterns that may need further investigation.
- Instruction:
  a. Calculate the Mean
  b. Calculate the Standard Deviation
  c. Calculate the Z-Score
  d. Highlight Outliers

	A	B
1	**Branch**	**Monthly Sales (RM)**
2	A	15000
3	B	16000
4	C	15200
5	D	15800
6	E	14900
7	F	15400
8	G	30000
9	H	15500
10	I	15300
11	J	15700

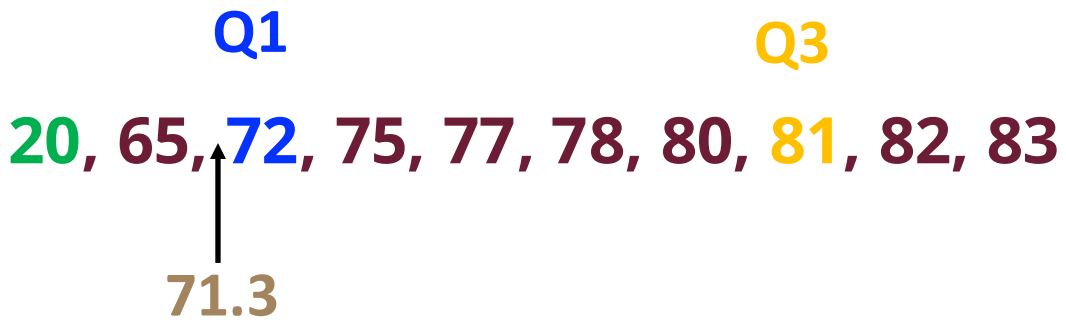
IQR Method (Interquartile Range)

- Quartiles break the date set into four equal portions.
- The first quartile, or the **Q1** , is the point below **25%** of the data.
- The second quartile, also known as the median or **Q2** , is the midpoint of the data.
- The third quartile, or **Q3** , is the point where **75%** of the data is below it.
- Sitting between Q1 and Q3 , the **interquartile range** (IQR = **Q3** - **Q1**) is used in calculating outliers.



Example of Outliers

- Data set student marks: 20, 65, 72, 75, 77, 78, 80, 81, 82, 83.
- Average = 713/10 = **71.3**



- In this case, the Q1 is 72, and the Q3 is 81; therefore, the IQR is 9 [81-72].
- The **lower bound** is defined as Q1 (72) minus 1.5 multiplied by IQR, resulting in 58.5 [72-13.5=58.5].
- The **upper bound** is calculated by adding the result of 1.5 times the IQR to the Q3 , resulting in 94.5 [9x1.5 = 13.5; 81+13.5 = 94.5].
- A score outside of this range is considered an outlier. As you can see here, the value of **20** is an outlier since it is below the lower fence.



CLASS ACTIVITY

Scenario: Monthly Sales of a Store

- Let's say a store records the number of **units sold each month**:

Month	Units Sold
January	120
February	115
March	118
April	121
May	500
June	119
July	117
August	116

Instruction:

1. Identify Q1,Q3 and IQR
2. Lower Bound
3. Upper Bound
4. Outliers

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Exercise Excel Functions

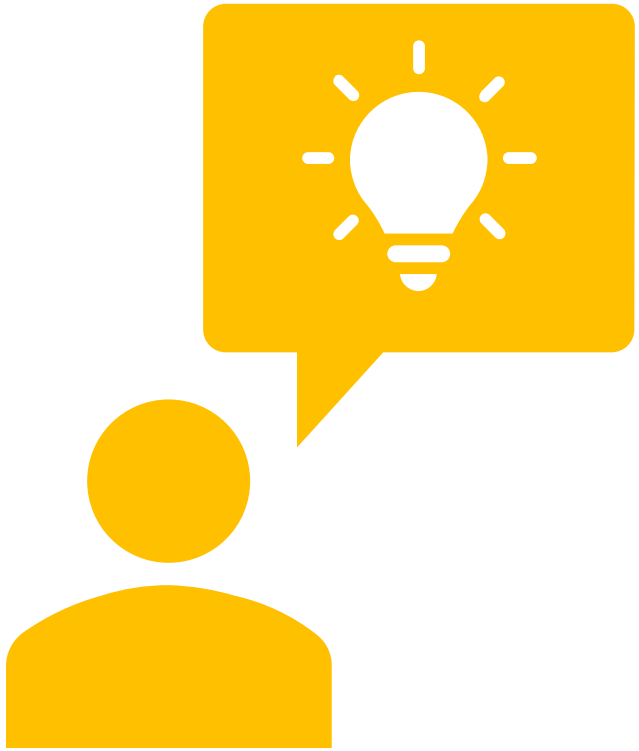
Scenario: Daily Website Visitors

- Let's say a blog tracks the number of **visitors per day** for 10 days, identify:
  a. Q1
  b. Q3
  c. IQR
  d. Lower Bound
  e. Upper Bound
  f. Using IF statement identify data is normal or outliers

Data

	A	B
1	**Day**	**Visitors**
2	Day 1	150
3	Day 2	152
4	Day 3	148
5	Day 4	153
6	Day 5	149
7	Day 6	151
8	Day 7	400
9	Day 8	150
10	Day 9	147
11	Day 10	149

KNOWLEDGE CHECK



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